Carmel Holy Word Secondary School
2024 – 2025 Term 2 Exam

Secondary 3 Computer Literacy

Question Book

Date	: 13th June 2025
Time	: 10:40 a.m. – 11:10 a.m.
Time Limit	: 30 minutes
Pages	: 7
Total mark	: 40

Instructions:

1. You should write your name, class, class number and the examination venue in the spaces provided on Page 1 after the examination start announcement is made.

2. There are three sections, A, B and C, in this Paper.

3. **ANSWER ALL QUESTIONS**. Answers to Section A, B and C should be written in the answer sheet provided.

Name	
Class	
Class No.	
Venue	

By Markers

Total	/ 40

Section A: Multiple Choice Questions (15 marks)
Choose the most suitable answers.

1. What is the output of the following program?

```
print("I Love \"Jesus\"")
```

A. `I Love \"Jesus\"`
B. `I Love "Jesus"`
C. `I Love 'Jesus'`
D. This program cannot be compiled.

2. Which of the following codes can be compiled successfully?

A. `1Love = Jesus`
B. `2Square = 4`
C. `_3_3 = 9`
D. `import = "3E"`

3. What is the output of the following program?

```
i = "3"
j = int(i + i)
print(str(j) + str(j))
```

A. `3333`
B. `66`
C. `12`
D. `9`

4. Read the following program.

```
L = input()
R = input()
print(R + " in the " + L + " always!")
```

Input the following text in the program in order:

```
Lord
Rejoice
```

What is the output of the program in the terminal?

A. `R in the L always!`
B. `Rejoice in the Lord always!`
C. `Lord in the Rejoice always!`
D. `R + in the + L + always!`

5. Read the following program.

```
k = int(input())
print(str(k + k) + str(k * k))
```

Input the following text in the program:

```
2
```

What is the output of the program in the terminal?

A. `224`
B. `44`
C. `2222`
D. `8`

6. Which of the following codes print text `Y2025` in the terminal?

A.
```
X = "Y"
print(X + str(20 + 25))
```
B.
```
X = 20
print("Y" + str(X * 101) + 5)
```
C.
```
X = "20"
print("Y" + X + str(int(X) + 5))
```
D.
```
X = "2025"
print("Y" + int(X))
```

7. Read the following program.

```
A = input()
B = A + input()
C = B + input()
print(A + B + C)
```

Input the following text in the program in order:

```
C
H
W
```

What is the output of the program in the terminal?

A. `CHWCHW`
B. `CCHHWW`
C. `CCHCHW`
D. `CCCHHW`

8. Read the following program.

```
A = int(input())
B = int(input())
print(A % 4 + B)
```

Input the following text in the program in order:

```
34
33
```

What is the output of the program in the terminal?

A. 32
B. 33
C. 34
D. 35

9. What is the output of the following program?

```
x = 101
if x >= 100:
    x = x - 1
elif x <= 100:
    x = x - 2
print(x)
```

A. 97
B. 98
C. 99
D. 100

10. What is the output of the following program?

```
x = 4
while x <= 100:
    x = x + x
print(x)
```

A. 64
B. 128
C. 104
D. This program cannot be compiled

11. What does the following code output?

```
import random
print(random.randint(1, 3))
```

A. Always prints 1
B. Always prints 3
C. Prints a random integer: 1, 2, or 3
D. Error because `randint()` cannot be used

12. What library does the following code require to work?

```
cv2.imshow("Frame", frame)
```

A. `gTTS`
B. `OpenCV`
C. `GoogleCloudTranslate`
D. `SpeechRecognition`

13. What is the purpose of `cv2.waitKey(0)` in a video loop?

A. Waits for a key press
B. Displays audio input
C. Converts speech to text
D. Saves an image

14. Which statement about `gTTS` is **not** correct?

A. It converts text to mp3 audio
B. It needs internet access
C. It requires `playsound` to play audio
D. It can translate text into Chinese

15. Which of the following lines records the user's voice using the `speech_recognition` module?

A. `text = recognizer.recognize_google(audio)`
B. `tts = gTTS("Hello")`
C. `cv2.imread("image.jpg")`
D. `result = translator.translate("Hello", dest="zh-tw")`

Section B: Short Questions (15 marks)
All answers should be written in the spaces provided in the answer sheet.

1. Fill in the blanks to complete the text-to-speech program. (5 marks)

gtts	gTTS	love.mp3	save	playsound	input

```
from ____(i)_____ import gTTS
from ____(ii)______ import playsound
tts = ____(iii)____("Jesus loves you", lang="en")
tts.___(iv)_____("love.mp3")
playsound("____(v)____")
```

2. Fill in the blanks to translate the English sentence "God loves you" into Chinese. (5 marks)

en	input	Client	zh	translate	result

```
from google.cloud import translate_v2 as ____(i)_____
import os
os.environ["GOOGLE_APPLICATION_CREDENTIALS"] = "password.json"
client = translate. ____(ii)_____ ()
result = client.translate("God loves you",
    source_language="____(iii)_____",
    target_language="____(iv)_____")
print(____(v)_____["translatedText"])
```

3. Fill in the blanks to complete the speech recognition program. (5 marks)

source	recognizer	sr	text	translate	audio

```
import speech_recognition as ____(i)_____
recognizer = sr.Recognizer()
with sr.Microphone() as source:
   audio = ____(ii)_____.listen(____(iii)_____)
   text = recognizer.recognize_google(____(iv)_____, language="en")
   print(____(v)_____)
```

Section C: Long Question (10 marks)

1. A student wants to write a Python program using OpenCV to load a video and detect faces with a pre-trained Haar Cascade model.

   Below is the incomplete code:

```
Line  Program
 1    import cv2
 2
 3    # Load the Haar Cascade face detection model
 4    face_cascade =
 5    cv2.CascadeClassifier("haarcascade_frontalface.xml")
 6
 7    # Load video
 8    video = cv2.VideoCapture('WalkingGirl.mp4')
 9
10    while True:
11        ret, frame = video.read()
12        if not ret:
13           break
14
15        gray = cv2.cvtColor(frame, cv2.COLOR_BGR2GRAY)
16        faces = face_cascade.detectMultiScale(gray, 1.1, 4)
17
18        for (x, y, w, h) in faces:
19           cv2.rectangle(frame, (x, y), (x+w, y+h), (255, 0, 0), 2)
20
21        cv2.imshow("Face Detection", frame)
22        cv2.waitKey(1)
```

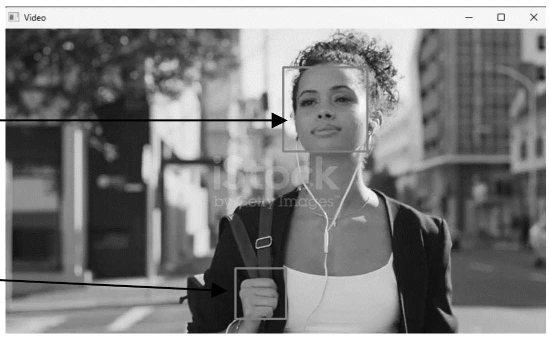
(a) Which two files are required for running this program. (2 marks)

(b) Describe the purpose of line 12 and 13. (2 marks)

(c) What is the meaning of `x`, `y`, `w` and `h` in line 18 and 19? (4 marks)

(d) The student observed that some detected objects in certain frames are not actual faces.

Face detected correctly



Incorrectly detected

(i) Give a reason for this problem. (1 mark)

(ii) Which line of code should be modified to fix this issue? (1 mark)

End of Paper

Carmel Holy Word Secondary School
2nd Term Exam 2024 - 2025
Secondary 3 Computer Literacy
Answer Sheet

/40

Name : ___________________ Time allowed : 30 minutes

Section A: Multiple Choice Questions (15 marks)

Choose the most suitable answers. Use an **HB pencil** to blacken the circle next to the question numbers below.

Class | Class No.

ZIPGRADE.COM

1	A	B	C	D	13	A	B	C	D
2	A	B	C	D	14	A	B	C	D
3	A	B	C	D	15	A	B	C	D
4	A	B	C	D					
5	A	B	C	D					
6	A	B	C	D					
7	A	B	C	D					
8	A	B	C	D					
9	A	B	C	D					
10	A	B	C	D					
11	A	B	C	D					
12	A	B	C	D					

15-Q (2574)

Section B: Short Questions (15 marks)

All answers should be written in the spaces provided.

1.

(i)	(ii)	(iii)
(iv)	(v)	

2.

(i)	(ii)	(iii)
(iv)	(v)	

3.

(i)	(ii)	(iii)
(iv)	(v)	

Section C: Long Question (10 marks)

1. (a) File 1: ______________________________

File 2: ______________________________

(b) ______________________________

____

(c) `x:` ____________ of detected faces.

`y:` ____________ of detected faces.

`w:` ____________ of detected faces.

`h:` ____________ of detected faces.

(d) (i) ______________________________

____

(ii) Line: ____________

End of Paper

2024-2025 Term 1 S3 CMP Examination
Suggested Answer

Section A: @ 1mark
BCABB CCDDB CBADA

Section B:

1.

(i) **`gtts`**	(ii) **`playsound`**	(iii) **`gTTS`**
(iv) **`save`**	(v) **`love.mp3`**	

2.

(i) **`translate`**	(ii) **`Client`**	(iii) **`en`**
(iv) **`zh`**	(v) **`result`**	

3.

(i) **`sr`**	(ii) **`recognizer`**	(iii) **`source`**
(iv) **`audio`**	(v) **`text`**	

Section C:

1. (a) File 1: **haarcascade_frontalface.xml**

   File 2: **WalkingGirl.mp4**

   (b) **It quits the loop if no more frames are available.**

   (c) `x`: **x-coordinate** of detected faces.

   `y`: **y-coordinate** of detected faces.

   `w`: **width** of detected faces.

   `h`: **height** of detected faces.

   (d) (i) **This happens because the program wrongly sees non-faces as faces.**

   (ii) Line: **16**